

Quinn M. Gallagher

Ph.D. Candidate, Princeton University

E-Quad A108, Princeton University, Princeton, NJ 08540

qg1361@princeton.edu | (610) 505-5458 | [Google Scholar](#) | [Personal Website](#)

EDUCATION

Princeton University

Ph.D. Candidate in Chemical and Biological Engineering

Advisor: Michael A. Webb

Princeton, NJ

Aug. 2022 – Present

University of Pennsylvania

B.S.E. in Chemical and Biomolecular Engineering, *summa cum laude*

Philadelphia, PA

Aug. 2018 – May 2022

SKILLS

Molecular dynamics (LAMMPS, OpenMM, Gromacs), machine learning (scikit-learn, PyTorch), active learning, Bayesian optimization (GPyTorch, GPy), machine learning force fields (TorchANI, UF3), density functional theory (Psi4, ORCA)

HONORS AND AWARDS

AI for Accelerating Invention Graduate Fellow , Princeton University	2025
William R. Schowalter Travel Grant , Princeton University	2025, 2026
Excellence in Teaching Award , Princeton University	2024
Francis Robbins Upton Fellowship , Princeton University	2022
NSF Graduate Research Fellowship , National Science Foundation	2022
Hugh Otto Wolf Memorial Prize , University of Pennsylvania	2022
Future Leader in Chemical Engineering , North Carolina State University	2021
Penn Engineering Exceptional Service Award , University of Pennsylvania	2021
Tau Beta Pi , University of Pennsylvania	2021
Manfred Altman Memorial Award , University of Pennsylvania	2020
Holtz Award , University of Pennsylvania	2019

PUBLICATIONS

8. Minimizing Data Requirements for Material Property Prediction via Cluster-based Training Set Selection
Q.M. Gallagher, S. Hasko, M.A. Webb, *ChemRxiv* (2026). Submitted. [\[Preprint\]](#)
7. A Local Structural Basis to Resolve Amorphous Ices
Q.M. Gallagher*, R.J. Szukalo*, N. Giovambattista, P.G. Debenedetti, M.A. Webb, *arXiv* (2026). Submitted.
[\[Preprint\]](#) *Equal contribution
6. A User's Guide to Your First Self-Driving Liquid Handling Lab
A. Maroulis, D. Waynor, **Q.M. Gallagher**, R.A. Patel, M. Tamasi, D.C. Radford, M.A. Webb, A.J. Gormley, *Digital Discovery* (2026). [\[Link\]](#)
5. Pressure-Consistent Iterative Boltzmann Inversion for Coarse-Grained Molecular Dynamics
Z. Yu, R.J. Szukalo, **Q.M. Gallagher**, M.A. Webb, *Journal of Chemical Theory and Computation*, 21, 20 (2025).
[\[Link\]](#)
4. Sustainable Recovery of Perfluoroalkyl Acids Using a Reusable Molecular Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, M.A. Webb, A. Criado, J. Mosquera, *Journal of Materials Chemistry A*, 13, 30 (2025). [\[Link\]](#)
3. Bayesian Optimization Hackathon for Chemistry and Materials
S. Baird, ..., **Q.M. Gallagher**, ..., Y. Zo, *ChemRxiv* (2025). [\[Preprint\]](#)

2. Data Efficiency of Classification Strategies for Chemical and Materials Design
Q.M. Gallagher, M.A. Webb, *Digital Discovery*, 4, 1 (2025). [\[Link\]](#)
1. Engineering a Surfactant Trap via Postassembly Modification of an Imine Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, A.B. León, M.A. Webb, A. Criado, J. Mosquera, *Chemistry of Materials*, 36, 18 (2024). [\[Link\]](#)

TEACHING

Princeton University

Guest Lecturer for CBE 512: Machine Learning for Chemical Science and Engineering October 2025
Teaching Assistant for CBE 250: Separations in Chemical Engineering Aug. 2024 – Dec. 2024
Teaching Assistant for CBE 503: Advanced Thermodynamics Aug. 2023 – Dec. 2023
Received the Excellence in Teaching Award from Princeton's School of Engineering and Applied Science.

University of Pennsylvania

Teaching Assistant for CIS 160: Mathematical Foundations of Computer Science Jan. 2020 – May 2022
Teacher for Access Engineering Jan. 2019 – May 2022

MENTORING

Undergraduate Students:

- Khang Tran, AI Fluency Program (March 2026 –)
- Sonia Hasko, CBE (Sept. 2024 – May 2025)
- David Opong, CBE (Sept. 2023 – May 2024)

High School Students:

- Michelle Gao, Laboratory Learning Program (May 2024 – Aug. 2024)

ACADEMIC SERVICE

Journal Reviewer: *ACS Macro Letters*

Professional Membership: AIChE, ACS

OUTREACH

Graduate Fellow, AI Lab, AI for Accelerating Invention Initiative, Princeton University Jan. 2026 –
Reviewer, NSF GRFP Workshop for CBE Students, Princeton University Sept. 2022 –
President, Access Engineering, University of Pennsylvania Sept. 2018 – May 2022
Vice President, Student Chapter of AIChE, University of Pennsylvania Sept. 2018 – May 2022
Secretary, Engineering Deans' Advisory Board, University of Pennsylvania Sept. 2018 – May 2022

PRESENTATIONS

11. "Minimizing Data Requirements for Material Property Prediction *via* Cluster-based Training Set Selection" (*Poster*) **Gordon Research Conference**, *AI for Materials, Energy and Chemical Sciences*, 2026.
10. "Automatic Construction of Data-Driven Collective Variables Using Local Atomic Environments" (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Machine Learning for Soft and Hard Materials II: Hard Matter and Methods*, 2025.
9. "Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships" (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Automated Molecular and Materials Discovery: Integrating Machine Learning, Simulation, and Experiment*, 2025.

8. “Data-driven Order Parameters for Atomic Systems: A Case Study on Amorphous Ices” (*Talk*) **Princeton CBE Graduate Student Symposium**, 2025.
7. “An Interpretable Machine Learning Tool for the Classification of Atomic States in Molecular Simulations” (*Poster*) **6th Molecular Simulations Workshop (NJIT)**, 2025.
6. “Minimizing Data Requirements for Material Property Prediction” (*Poster*) **AutoML Center for Accelerated Formulations Engineering Workshop**, 2025.
5. “Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships” (*Poster*) **Accelerate Conference**, 2025.
4. “Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships” (*Poster*) **New York City AI4Chemistry Summit**, 2025.
3. “Data-driven identification of collective variables for transitions between high-density amorphous and low-density amorphous ice” (*Talk*) **American Chemical Society, Middle Atlantic Regional Meeting**, 2025.
2. “Fewer Experiments, More Knowledge: Data-Efficient Machine Learning for Materials Design” (*Poster*) **Princeton CBE Graduate Student Symposium**, 2024.
1. “Modulating Heterochromatin Formation for Direct Cell Reprogramming” (*Talk*) **North Carolina State University, Future Leaders in Chemical Engineering National Award Symposium**, 2021.